BEI ZHANG

M.S. Candidate in Geography | Earth System Modeling, Atmospheric Aerosols and Climate
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RESEARCH PROFILE

Research interests: Earth system and climate modeling, aerosol microphysics and new particle formation, aerosol cloud climate interactions, scientific computing, model development, and the use of machine learning in atmospheric and climate modeling.

Core strengths: CESM and Fortran model development, aerosol microphysics and nucleation implementation, tracer and aerosol coupling, numerical debugging, HPC scientific computing, and observation based model evaluation.

EDUCATION

M.S. in Geography, Tianjin University Sep 2024 – Jun 2027 (expected)
School of Earth System Science | GPA: 3.9/4.0 | Ranked **1st/46** in major

B.S. in Geography, Tianjin University Sep 2020 – Jun 2024
Minor in **Philosophy** | GPA: 3.75/4.0 | Top 20%

PUBLICATIONS

1. **Zhang, B.**, Guo, Q., Sun, Y., Cui, L., & Zhu, J. (2026). Global modeling of atmospheric amines: Sources and sinks. ***Journal of Geophysical Research: Atmospheres,*** *131*(15), e2025JD045898. https://doi.org/10.1029/2025JD045898. **First author.**
2. Sun, Y., **Zhang, B.**, Zhao, X., & Zhu, J. (2025). Effects of clean fuel application on the radiative impact of ship sourced sulfate aerosols [in Chinese]. ***Earth Science Frontiers****, 32*(3), 263–273. **Second author.**
3. Zhao, J., Chen, W., Liu, Z., Liu, W., Li, K., **Zhang, B.**, Zhang, Y., Yu, L., & Sakai, T. (2024). Urban expansion, economic development, and carbon emissions: Trends, patterns, and decoupling in mainland China's provincial capitals (1985–2020). ***Ecological Indicators****, 169*, 112777. https://doi.org/10.1016/j.ecolind.2024.112777. **Co-author.**
4. Cui, L., **Zhang, B.**, Guo, Q., & Zhu, J. (2026). An interactive marine biogenic volatile organic compound emissions module in an Earth system model. ***Journal of Advances in Modeling Earth Systems***. Manuscript under first round revision; **Co-author**.
5. Ammonia emissions from penguin colonies mitigate warming in Antarctica. ***PNAS***. Manuscript under review. **Co-author.**

RESEARCH EXPERIENCE

Master's Research (2 stages)

Stage1: Global Amine Emission Inventory and CESM Modeling Sep 2024 – Present
Advisor: Prof. Jialei Zhu, Tianjin University | Resulted in first-author JGR: Atmospheres publication

- Developed a **global emission inventory** for methylamine (MMA), dimethylamine (DMA), and trimethylamine (TMA), combining **six anthropogenic sectors** with a **marine biogenic source**. Built Python workflows for meteorological data processing, emission ratio calculations, and gridded inventory construction.
- Configured and ran CESM v1.2.2 with the IMPACT aerosol module, and implemented atmospheric amine processes including **OH oxidation, wet deposition, dry deposition, and heterogeneous aerosol uptake**.
- Evaluated model simulations against observations from 27 global sites, improved marine TMA representation through a new **ocean emission parameterization**, and quantified changes in **aerosol radiative forcing** associated with amines.
- Performed **source attribution, source sink budget analysis, and sensitivity simulations** to quantify the controls on global amine distributions, atmospheric lifetimes, and regional burden patterns.
- Led the **model analysis, uncertainty assessment, manuscript writing, and revision** for the resulting first author paper.

Stage2: CESM Model Development: Coupled New Particle Formation Module Feb 2026 – Present
Second stage of thesis work; manuscript in preparation

- Built and implemented a **multi mechanism new particle formation framework** in CESM 1.2.2 by adapting CMAN from E3SM. Developed **Fortran interfaces** connecting sulfur chemistry and aerosol microphysics, including species and tracer mapping, unit conversion, and array mapping between the two model architectures.
- Integrated **amine and sulfuric acid nucleation**, precursor consumption, particle number and mass allocation, sulfate and organic aerosol routing, coagulation, and modal transfer. The framework now **compiles and runs successfully**.
- Extended the framework to represent **amine driven nucleation** involving MMA, DMA, and TMA, using ACDC based parameterizations and literature constraints. Evaluated nucleation behaviours across sulfuric acid, amine concentration, and temperature ranges to assess the physical consistency of the implemented mechanisms.
- Conducting model validation through **source code diagnostics and controlled sensitivity simulations**, focusing on mass and number conservation, unit consistency, aerosol mode logic, precursor depletion, coagulation, and process routing. Current work is identifying the sources of unexpected aerosol responses and refining the coupled framework.

Urban Expansion and Carbon-Emission Decoupling — B.S. Thesis Sep 2023 – Jun 2024

- Processed **1985–2020 impervious-surface and socioeconomic datasets** in ArcGIS and contributed spatial analysis and visualization for 31 provincial capitals in mainland China.
- Applied **carbon emission decomposition** and **elasticity based decoupling** analysis to investigate long-term links among urban expansion, economic growth, and carbon emissions; work published in Ecological Indicators (2024).

China Postgraduate Mathematical Contest in Modeling 2024 & 2025

National Second Prize in two consecutive years

- Developed reproducible **Python workflows** for large heterogeneous datasets under five day competition deadlines, covering data preprocessing, feature engineering, model development, validation, visualization, and scientific reporting.
- 2024: Processed 0.25° gridded precipitation data together with terrain and temperature datasets. Implemented coordinate alignment, resampling, normalization, temporal feature construction, and model comparison using **Random Forest, ARIMA, and LSTM**, with time series cross validation to assess predictive performance.
- 2025: Engineered **29 time, frequency, and wavelet domain features** from vibration signals and built a **PyTorch 1D CNN / DANN workflow** for transfer learning. Integrated sensor fusion, distribution-based sample selection, and **t SNE diagnostics** to evaluate feature alignment between source and target domains.

National Key R&D Program of China — Research Participant 2023 – 2025

Project: Anthropogenic organic compounds, new particle formation, and climate effects in megacities (2022YFE0135000).

PROFESSIONAL EXPERIENCE

Research Intern, Policy and Programme Division, UN Environment Programme (UNEP HQ) Sep 2026 – Nov 2026

Strategic Planning, Reporting and Impact Measurement | ***Nairobi, Kenya*** | *Air quality impact measurement*

- Supported UNEP-wide impact-measurement work by screening closed-project portfolios, cross-validating project metadata, and identifying air-quality cases suitable for quasi-experimental evaluation using institutional data systems.
- Developed and compared causal-inference designs for environmental policy evaluation, including interrupted time series, synthetic control, spatial regression discontinuity, matching, and difference-in-differences; translated methodological requirements and data needs into practical guidance for programme teams.
- Contributed to research-design documentation and cross-divisional workshops linking impact pathways, external datasets, counterfactual methods, and organization-level environmental indicators.

ADDITIONAL EXPERIENCE

Micro-degree in Global Governance, Ministry of Education Program Oct 2025 – May 2027

- Training in UN operational mechanisms, systems thinking, and cross-cultural management.

Debate Team Leader, Tianjin University Jun 2024 – Jun 2025

- Led a 15-member team in competitions; strengthened structured argumentation and scientific communication skills.

University-level Undergraduate Independent Research Project Jun 2023 – Jun 2024

- Led proposal development, mid-term and final project defenses, and project-fund administration for a university-level independent research project.

SELECTED HONORS & AWARDS

National Second Prize (two consecutive years), China Postgraduate Mathematical Contest in Modeling	2024, 2025
First-Class Scholarship, School of Earth System Science, Tianjin University	2024, 2025, 2026
Outstanding Class Officer (University Level), Tianjin University	2025
Ethnic Role Model Scholarship (University Level), Tianjin University	2025

TECHNICAL SKILLS

Earth System Modeling: CESM; IMPACT aerosol module; atmospheric chemistry and aerosol-process implementation
Programming: Fortran, Python (pandas, NumPy, scikit learn, PyTorch), MATLAB; Linux/HPC.
Model Development: Fortran source code modification, process coupling, tracer and aerosol routing, unit conversion, model diagnostics, and numerical debugging in CESM.
Data Analysis: Machine Learning; NetCDF/climate-data processing, advanced statistics (including causal inference)
Geospatial Evaluation: model evaluation, ArcGIS, ENVI; scientific visualization and reproducible data workflows.
Languages: Chinese (native); English (IELTS 6.5, CET-6, professional working experience in an international environment)